



# In Silico Genome-Wide Mining and Analysis of Terpene Synthase Gene Family in *Hevea Brasiliensis*

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Received: 18 September 2022 / Accepted: 9 December 2022 / Published online: 19 December 2022  
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## Abstract

Terpene synthases (TPSs) catalyze terpenoid synthesis and affect the intracellular isopentenyl diphosphate (IPP) and dimethylallyl diphosphate (DMAPP) concentration. In this study, we mined the in silico genome-wide TPS genes of *Hevea brasiliensis* and identified 47 full-length TPS genes. They had DDXXD, DXDD, NSE/DTE, RR(X)8W, EA(X)W, and other conserved motifs. The phylogenetic tree analysis revealed that the TPSs of *H.brasiliensis* (HbTPSs) were divided into five subfamilies, TPS-a, TPS-b, TPS-c, TPS-e/f, and TPS-g. HbTPSs were predicted to have functions in the cellular components, molecular functions, and biological processes. HbTPSs were involved in seven pathways, which were K14173, K14175, K15803, K04120, K04121, K17982, and K12742 in the secondary metabolite pathway prediction. Three-dimensional structures of HbTPSs of 7 pathways were predicted, and DDXXD, NSE/DTE, and EA(X)W conserved motifs near the binding sites were found. Cis-acting elements analysis showed that they had more cis-acting elements related to phytohormone responsiveness, which indicated that terpenoid biosynthesis might be related to phytohormone regulation. RNA-Seq analysis showed that different HbTPSs were expressed differentially in different tissues. This study's results help reveal the role of HbTPSs and their molecular mechanism and help resolve the regulatory mechanism of terpenoid biosynthesis in *H.brasiliensis*.

**Keywords** In silico · Genome-wide · *Hevea brasiliensis* · Terpene synthase · Mining · Analysis

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## Introduction

Terpenoids are a specific class of plant metabolites that play a significant role in plant growth, development, and the interaction between plants and the environment (Tholl 2015). To date, more than 95,000 terpenoids have been identified (Faylo et al. 2021). Although terpenoids are very diverse, they are all derived from the biosynthesis of the common precursors isopentenyl diphosphate (IPP) and dimethylallyl diphosphate (DMAPP) (Newman and Chappell 1999). There are two main synthetic pathways, the methylerythritol phosphate pathway (MEP) in the plastid and the mevalonate pathway (MVA) in the cytoplasm (Vranová et al. 2013). IPP and DMAPP are condensed by isoprene transferase to produce the precursors geranyl diphosphate (GPP, C<sub>10</sub>), farnesyl diphosphate (FPP, C<sub>15</sub>), and geranylgeranyl diphosphate (GGPP, C<sub>20</sub>) of most terpenoids. Subsequently, terpene synthases (TPSs) and isoprene transferases catalyze the precursors to form a variety of terpenoids, including hemiterpenes (C<sub>5</sub>), monoterpenes (C<sub>10</sub>), sesquiterpenes (C<sub>15</sub>), diterpenes (C<sub>20</sub>), and polyterpenes (McGarvey and Croteau 1995; Tholl 2006; Yamada et al. 2015). In the terpene biosynthetic pathways, TPSs are located at the immediate downstream of the branch and are critical enzymes in terpene biosynthesis, which affect the concentration and flow of intracellular terpene skeletal monomer IPP and DMAPP (Bohlmann et al. 1998b; Sharkey et al. 2013; Alquézar et al. 2017). Therefore, the analysis of TPS's structure, chemical components, molecular functions, and secondary metabolite pathways is related to the intercellular concentration of terpene skeletal monomer and can help to resolve the regulation of biosynthesis of terpenoids.

Based on the catalytic mechanism of TPSs, TPSs are usually classified into two main classes: class I TPSs use trinuclear metal clusters to trigger the ionization of isoprenoid diphosphate substrates to yield allylic cations and inorganic pyrophosphates, while class II TPSs rely on general acids (aspartic acid side chains) of the carbon–carbon double bonds of isoprenoid substrates to yield tertiary carbocations (Hasserodt et al. 1997, 2000; Wendt and Schulz 1998; Chiana et al. 1999). According to the sequence similarity of TPSs, TPSs can be classified into seven subfamilies, namely TPS-a, TPS-b, TPS-c, TPS-d, TPS-e/f, TPS-g, and TPS-h. Different subfamilies of TPSs can catalyze and produce different terpenoids.

There are also some conserved motifs in TPS genes, such as RR(X)8 W, DDXXD, NSE/DTE, RDR, DXDD, etc. (Chen et al. 2011), which have essential functions in catalytic reactions. For example, the RR(X)8 W motif plays a role in the initiation of isomerization cyclization reactions, the DDXXD motif is involved in the coordination of divalent metal ions and water molecules as well as the stabilization of the active site, and the DXDD motif is essential for the protonation initiation, etc. (Williams et al. 1998; Whittington et al. 2002; Prisic et al. 2007).

*Hevea brasiliensis* (the Brazilian rubber tree) is an important economic plant for its main secondary metabolites—natural rubber, which is a kind of polyterpene (Zwart et al. 2015; Wu et al. 2017). Natural rubber is widely used in industry, agriculture, national defense, transportation, medical and health care,

and daily life because of its excellent elasticity, good insulation, plasticity, as well as wear resistance, tensile strength, water, and gas barrier, etc. (Men et al. 2018). It is an indispensable strategic material and industrial raw material. 90% of the world's natural rubber requirements come from *H.brasiliensis* (Uthup et al. 2013). According to the International Rubber Study Group (IRSG), the global demand for natural rubber is expected to reach 14.37 million tons in 2022. As a result, many scientists focus on research on how to increase natural rubber production (Sando et al. 2009).

The biosynthetic pathways of natural rubber can be divided into four phases. The first phase is required to form the common C<sub>5</sub> structural units IPP and DMAPP. In the second phase, sequential condensation of IPP with DMAPP as primer substrates lead to the production of linear all-(E)- oligoprenyl diphosphates(C<sub>10-20</sub>). The third phase is characterized by sequential condensation of IPP with all-(E)-oligoprenyl diphosphate as primer substrate to form linear (Z, E)-mixed polyprenyl diphosphates. In the fourth phase, the (Z, E)-mixed polyprenyl diphosphates are modified, for example, by dephosphorylation of and reduction and transfer of the polyprenyl chain to other molecules (Men et al. 2018; Cherian et al. 2019; Yamashita and Takahashi 2020).

In addition to natural rubber, a polyterpene, other terpenes are present in *H.brasiliensis*. TPSs control the yield and kinds of these terpenes, which affect the concentration and flow of the terpene skeletal monomers IPP and DMAPP, which are also required for the biosynthesis of natural rubber. Although the genome of *H.brasiliensis* has been sequenced, no genome-wide TPS gene has been mined and analyzed so far. Therefore, in this study, all TPSs were excavated through the genome of *H.brasiliensis*. Gene sequence analysis, phylogenetic tree analysis, functional annotation, cis-acting element analysis, differential expression analysis, protein three-dimensional structure prediction and docking analysis with corresponding substrates were performed for HbTPSs. These analyses will help to reveal the role of HbTPSs and their molecular mechanism and provide new insights into the effective regulation of natural rubber biosynthesis in *H.brasiliensis*.

## Materials and Methods

### TPSs Mining and Identification

We downloaded genome sequences of *H.brasiliensis* (Rahman et al. 2013) and other species from the genome, a sub-database of the National Center for Biotechnology Information (NCBI) database (<https://www.ncbi.nlm.nih.gov/genome/>). Two conserved domain models of TPSs, namely the Terpene synthase N-terminal domain (PF01397) and the Terpene\_synth\_C (PF03936), were downloaded from the Pfam database (<http://pfam.xfam.org/>) (Finn et al. 2016). We searched their protein sequence data separately using HMMER 3.0 hmmsearch (Finn et al. 2011). The sequences containing one or two conserved domains were searched by filtering with the condition that the number of amino acid residues was greater than 200 and the

*E*-value was less than  $10^{-5}$ , and the protein sequences encoded by the same gene were taken as the longest one for analysis (Chaw et al. 2019; Chen et al. 2020).

### HbTPS Gene Sequences Analysis

For the mined HbTPSs, gene structure was analyzed based on the genome annotation information. The intron–exon structures of the HbTPS genes were shown using the Gene Structure Display Server 2.0 (Hu et al. 2015). Conserved motifs were analyzed using MEME (<https://meme-suite.org/meme/tools/meme>) (Bailey et al. 2015) and MUSCLE (Edgar 2004).

### Phylogenetic Tree Analysis of HbTPSs

MUSCLE was used for multiple sequence alignment and the phylogenetic tree was constructed using MEGA X (Sudhir et al. 2018) with parameters of neighbor-joining method, the Jones Taylor Thornton (JTT) model and the bootstrap value 1000. The phylogenetic tree results were further modified and embellished using the iTOL (<https://issstol.embl.de/>) (Ivica et al. 2019).

### Functional Annotation and Secondary Metabolite Pathway Prediction of HbTPSs

InterProScan was used to align the InterPro and Gene Ontology (GO) databases and obtain functions. The result was inferred in useful information on cellular components, molecular functions, metabolic processes, etc. The annotation results of the GO database were visualized using WEGO (<https://wego.genomics.cn/>) (Ye et al. 2006). The HbTPSs were mapped for annotation and prediction of putative functional genes involved in the biosynthesis of secondary metabolites against Kyoto Encyclopedia of Genes and Genomes (KEGG) metabolic pathways using KEGG Orthology And Links Annotation (BlastKOALA) (<http://www.kegg.jp/blastkoala/>) (Kanehisa et al. 2016). KOBAS (<http://kobas.cbi.pku.edu.cn/annotate/>) (Bu et al. 2021) was used to further complement the functional classification of HbTPSs.

### Three-Dimensional Structure Prediction and Molecular Docking Analysis of HbTPSs

We predicted the three-dimensional structure of HbTPSs using alphafold2 (<https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.ipynb>) (Jumper et al. 2021). The predictions were docked with their corresponding substrates using AutoDock (Seeliger and Groot 2010). The substrates farnesyl diphosphate (FPP), ent-copalyl diphosphate (ECP), geranylgeranyl diphosphate (GGPP), and dimethylallyl pyrophosphate (DMAPP) were downloaded in the Protein Data Bank database (<https://www1.rcsb.org/>) (Berman et al. 2000). The docking results were visualized using PyMol (DeLano 2002).

## Analysis of Cis-Acting Elements of HbTPS Genes

Based on the annotation information of HbTPSs, the 2000 bp sequence in the upstream of each HbTPS gene was extracted using TBtools (Chen et al. 2018), and cis-acting elements of HbTPSs were predicted using PlantCARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) (Lescot and Research 2002).

## Differential Expression Analysis of HbTPSs

The RNA-seq data measured in leaves and latexes of *H.brasiliensis* were downloaded from the sub-database SRA of the NCBI database (PRJNA742874) (Chen et al. 2021a). The data from samples SRR15012153, SRR15012154, and SRR15012155 were measured in latex, and data from samples SRR15012156, SRR15012157, and SRR15012158 sample were measured in leaves. RNA-seq data were downloaded and formatted using sra-tools, assessment of sequencing data quality was performed using fastqc, adaptors, and low-quality data were removed using trimmomatic (Bolger et al. 2014), RNA-seq datasets of *H.brasiliensis* were aligned to genomic sequences using Hisat2 (Kim et al. 2019), the sam datasets were sorted and formatted using samtools (Li et al. 2009), expression was calculated using featurecounts (Yang et al. 2013), and differential expression analysis was performed using DESeq2, samples from leaf were set as control (Love et al. 2014).

## Results

### TPSs Mining and Identification

Using HMMER software, the two conserved domains PF01397 and PF03936 of TPSs were used for similarity search in the protein sequences of *H.brasiliensis*. Among them, 63 sequences contained both the N-terminal domain and the C-terminal domain, and 12 sequences had only the C-terminal domain or the N-terminal domain. Among the 63 sequences, in the case of multiple protein sequences encoded by a single gene, only the longest one was taken for analysis; there were 13 such genes in total. There were 3 genes with a large number of uncertain nucleotide N that were not involved in the analysis. Finally, we obtained 47 relatively complete HbTPSs, and the information about these 47 HbTPS sequences was shown in supplementary Table S1 (refer to Fig. 3 for subfamily classification). All analyzed HbTPS gene sequences and coding sequences have been included in the supplemental folder Sequences files. The ids correspondence of HbTPSs was in supplementary Table S2.

### HbTPS Gene Sequences Analysis

The identification of terpenoid synthase genes and their coding sequence analysis were the basis of function, structure, and phylogenetic analysis, so we firstly



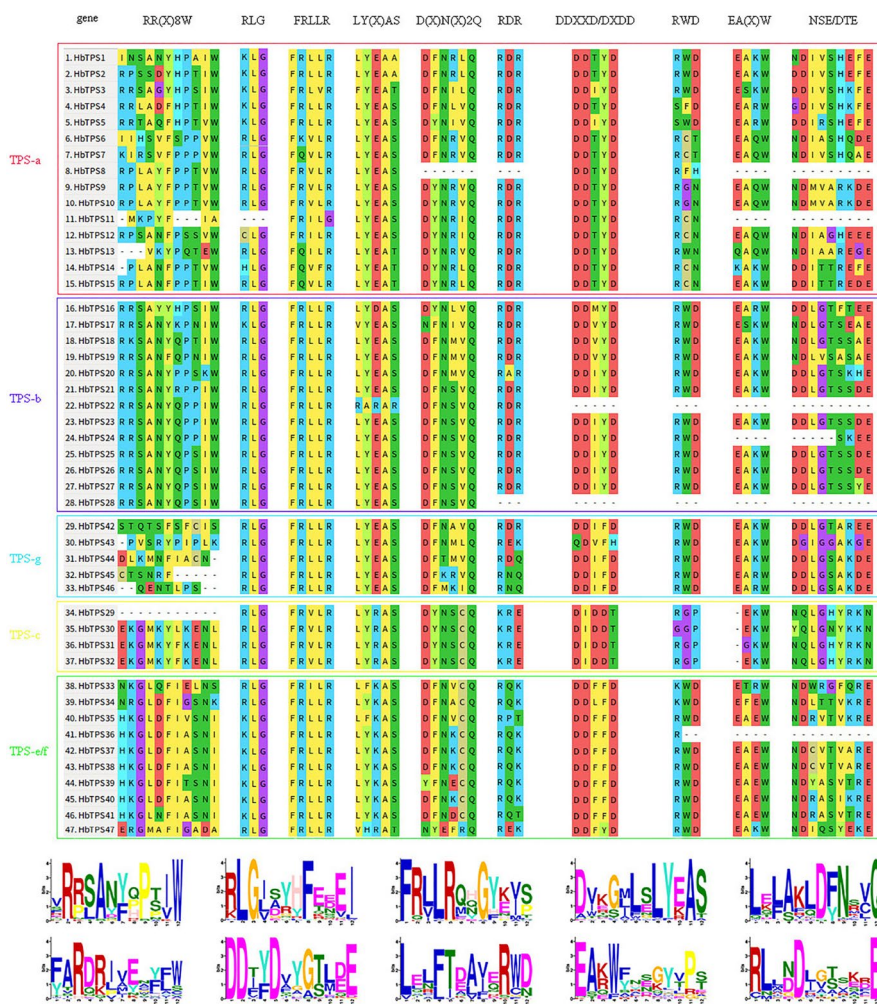
**Fig. 1** Exon–intron number of 47 HbTPS genes. The yellow rectangles represented exons, and the thin lines represented introns. Genes were shown in different colored boxes according to subfamily (Color figure online)

analyzed the sequences of identified terpenoid synthase genes. We showed the exon–intron numbers of HbTPS genes in Fig. 1 (refer to Fig. 3 for subfamily classification). Genes belonging to the TPS-a subfamily contained 4–8 exons and 3–7 introns, genes belonging to the TPS-b subfamily contained 3–8 exons and 2–7 introns, genes belonging to the TPS-c subfamily contained 10–15 exons and 9–14 introns, genes belonging to the TPS-e/f subfamily contained 8–15 exons and 7–14 introns, and genes belonging to the TPS-g subfamily contained 6–7 exons and 5–6 introns.

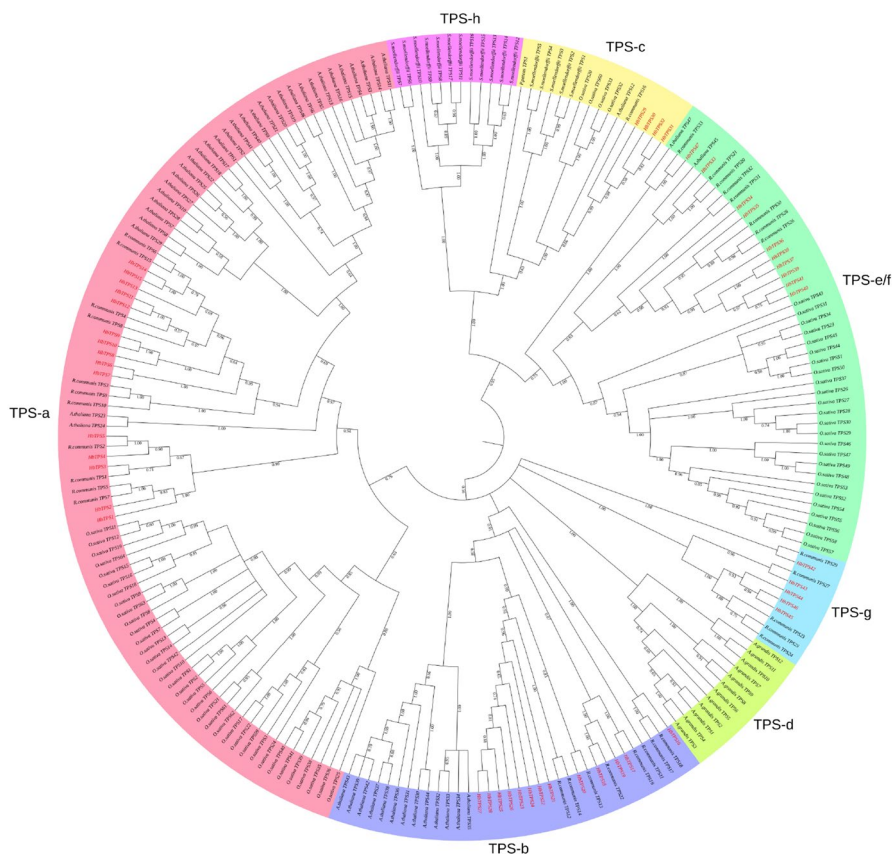
Multiple sequence alignment results revealed RR(X)8 W, RDR, DDXXD, DXDD, and NSE/DTE in HbTPSs, which were conserved motifs commonly found in other plants' TPSs. In addition, RLG, FRLLR, LY(X)AS, D(X)N(X)2Q, RWD,



and EA(X)W were also conserved motifs in HbTPSs. The RR(X)8 W motif was only present in TPS-a and TPS-b subfamilies and was very conserved in the TPS-b subfamily. The RLG, LY(X)AS, and D(X)N(X)2Q motifs were present in all TPS subfamilies of *H.brasiliensis*. The RDR motif was conserved in the TPS-a and TPS-b subfamilies and only in HbTPS42 of TPS-g subfamily. The FRLLR, DDXXD, EA(X)W, and NSE/DTE motifs were present in all subfamilies except the TPS-c subfamily (DXDD in the TPS-c subfamily shown in the same site of DDXXD). The RWD motif was relatively unconserved in the TPS-a subfamily, absent in the TPS-c subfamily, but conserved in the TPS-b and TPS-g subfamilies. Among all the motifs



**Fig. 2** Conserved motifs RR(X)8 W, RDR, DDXXD, DXDD, NSE/DTE, RLG, FRLLR, LY(X)AS, D(X)N(X)2Q, RWD, and EA(X)W in HbTPSs. Among them, RLG, FRLLR, LY(X)AS, D(X)N(X)2Q, RWD, and EA(X)W conserved motifs had been newly discovered in HbTPSs



**Fig. 3** Shown the results of the phylogenetic tree analysis of *H.brasiliensis* and other six species TPSs by neighborhood-joining method; the model was set to Jones Taylor Thornton, and the bootstrap value was set to 1000

found, they showed the highest conservation in the TPS-b subfamily and TPS-c subfamily contained the least conserved motifs. Conserved motifs were shown in Fig. 2.

### Phylogenetic Tree Analysis of HbTPSs

Gene sequence could reflect the evolution process of genes. The evolution processes of genes with similar sequences were also similar, so that they could be clustered into one class in molecular phylogenetic analysis. To analyze the evolution of HbTPSs, we selected TPSs from *Ricinus communis* (Euphorbiaceae), *Physcomitrella patens* (bryophytes), *Arabidopsis thaliana* (dicotyledons), *Selaginella moellendorffii* (ferns), *Oryza sativa* (monocotyledons), *Abies grandis* (gymnosperms), and HbTPSs for phylogenetic tree analysis. Except for *A.grandis*, which is a gymnosperm, from where 12 TPSs were found by blastp search in the NCBI database using the available sequence (Q9M7D1.1) belonging to the TPS-d subfamily, TPSs from all



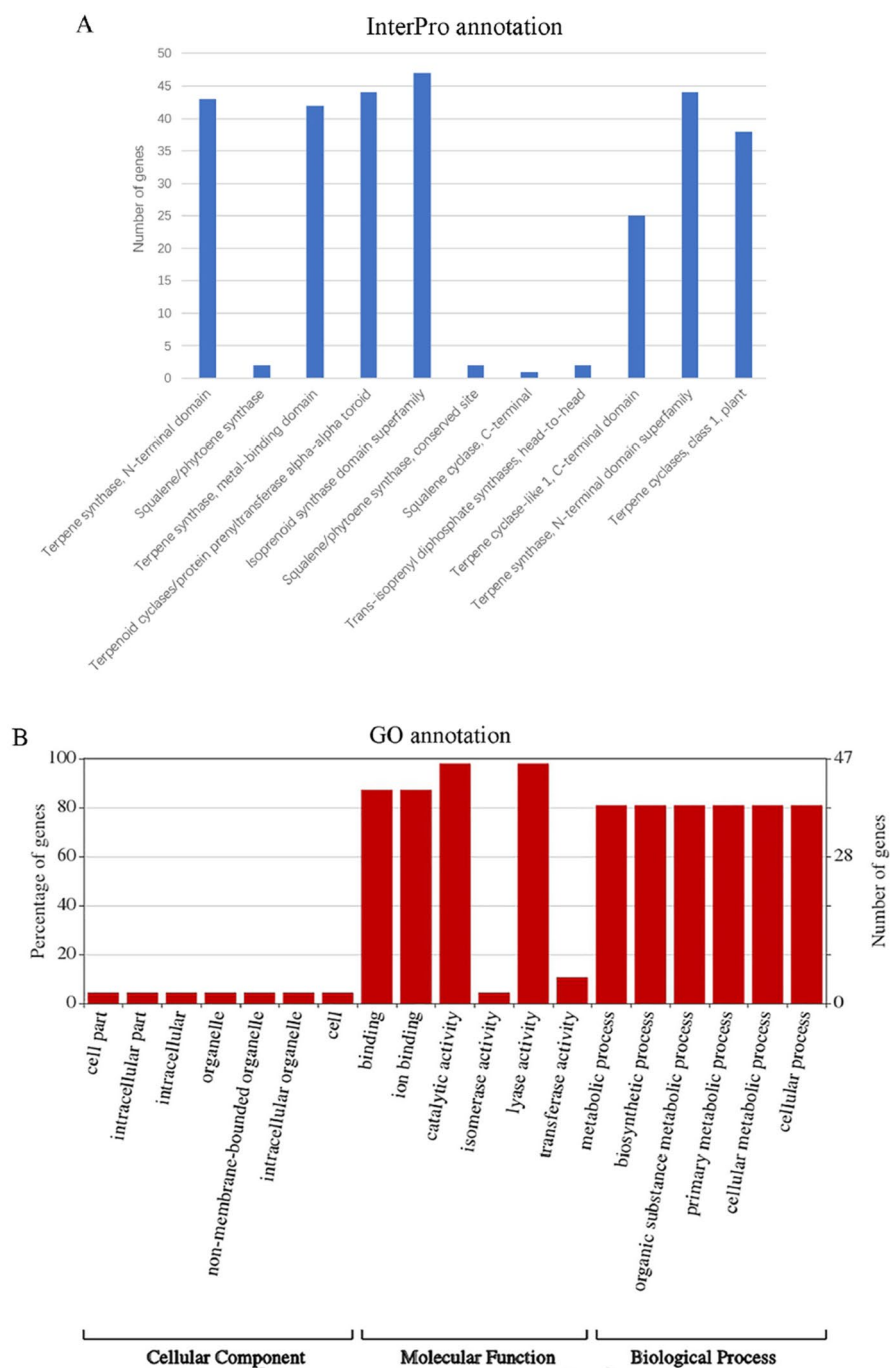
other five plants were identified by the same method as that used to identify TPSs of *H.brasiliensis*. The number of TPSs in the TPS subfamilies of *H.brasiliensis* and the other six plants was shown in supplementary Table S3. The TPS-a subfamily had the most significant number of TPSs, and the TPS-g subfamily had the least number of TPSs.

Results of the phylogenetic tree of TPSs of *H.brasiliensis* with six other plant species revealed (Fig. 3) that there were 15 HbTPSs in the TPS-a subfamily branch, from HbTPS1 to HbTPS15. The TPS-b subfamily branch had 13 HbTPSs, from HbTPS16 to HbTPS28. The TPS-c subfamily branch had 4 HbTPSs, from HbTPS29 to HbTPS32. The TPS-e/f subfamily branch had 10 HbTPSs, including HbTPS33 to HbTPS41 and HbTPS47. The TPS-g subfamily branch had 5 HbTPSs, from HbTPS42 to HbTPS46. It was found that the total number of HbTPSs in the TPS-a subfamily was the largest, which was 15. The number of HbTPSs in the TPS-a and TPS-b subfamilies reached 28, accounting for 59.57% of the total HbTPSs, and the TPS-c subfamily had the least HbTPSs with only 4. HbTPS did not appear in the TPS-d subfamily and TPS-h subfamily. HbTPSs clustered with TPSs of *R.communis* in all subfamilies. It was noteworthy that in the TPS-a subfamily, the TPSs of *H.brasiliensis*, *R.communis*, and *A.thaliana*, which belonged to dicotyledonous plants, clustered into one branch of the TPS-a subfamily, while the TPSs of *O.sativa*, which belonged to monocotyledonous plants, were clustered into another branch. TPS-h subfamily had only TPSs of *S.moellendorffii*, which was unique to *S.moellendorffii*.

## Functional Annotation and Pathway Mapping of HbTPSs

The proteins with similar sequences might have similar functions. We compared the sequences of the identified HbTPSs to the protein sequences in the existing databases (InterPro, GO and KEGG) to predict the functions of HbTPSs. The annotated results in the InterPro database show that all HbTPSs were associated with a total of 11 functions. All HbTPSs had IPR008949 (Isoprenoid synthase domain superfamily) function, HbTPS20 and HbTPS27 had IPR002060 (Squalene/phytoene synthase) function, HbTPS29 and HbTPS45 had IPR19845 (Squalene/phytoene synthase, consumed site) function, only HbTPS39 had IPR032696 (Squalene cyclase, C-terminal) function, IPR034741 (Terpene cyclase-like 1, C-terminal domain) function in 25 HbTPSs, IPR044814 (Terpene cyclases, class 1, plant) function in 38 HbTPSs, other functions existed in more than forty HbTPSs, the functions of these annotations were the functions of TPS. The annotation results were shown in Fig. 4A, and more detailed annotation information was presented in supplementary Table S1.

According to the annotation results of HbTPSs in the GO database, all annotation results were classified into three categories, Biological Process, Molecular Function, and Cellular Component. In Biological Process, six subclasses were involved, and 38 sequences were annotated to be involved in each subclass. In Molecular function, 46 sequences were annotated in lyase activity or catalytic activity, 41 sequences in binding or ion binding, 2 sequences in isomerase activity, and 5 sequences in



**Fig. 4** **A** Results of HbTPSs annotation in the InterPro database. **B** Results of HbTPSs annotation in the GO database

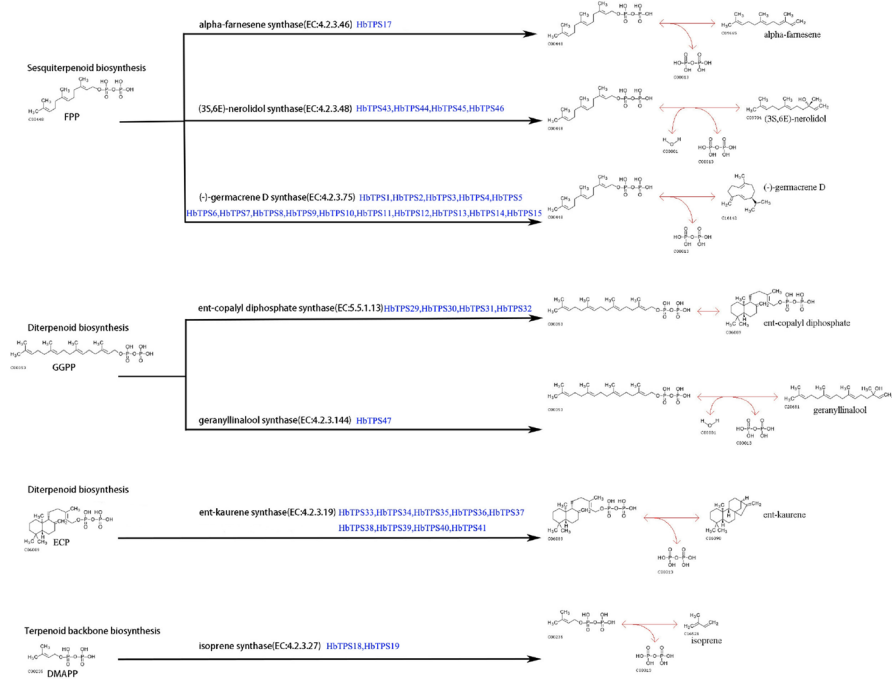
transferase activity. In Cellular Component, seven subclasses were involved and only two sequences were annotated in each subclass. There were the least HbTPSs annotated in the cellular component. The annotation results were shown in Fig. 4B, and more detailed information was shown in supplementary Table S1.

KEGG pathway mapping against the HbTPSs was performed to predict and hypothesize the secondary metabolite biosynthesis pathway. A total of 36 HbTPSs were successfully annotated, accounting for 76.60% of the total number. HbTPSs were predicted to be involved in seven secondary metabolite pathways, including sesquiterpenoid biosynthesis pathways K14173, K14175, and K15803, diterpenoid biosynthesis pathways K04120, K04121, and K17982, and terpenoid backbone biosynthesis pathway K12742. None of the HbTPS was predicted in the monoterpenoid biosynthesis pathway, but in the KOBAS database, HBTPS21, HBTPS23, HBTPS25, HBTPS26, and HBTPS27 were predicted as monoterpene synthases. According to the predicted results, 20 HbTPSs were predicted as sesquiterpene synthases, 14 HbTPSs were predicted as diterpene synthases, and 2 HbTPSs were predicted as isoprene synthases. HbTPS17 was predicted in the K14173 pathway (EC: 4.2.3.46), HbTPS43 to HbTPS46 were predicted in the K14175 pathway (EC: 4.2.3.48), HbTPS1 to HbTPS15 were predicted in the K15803 pathway (EC: 4.2.3.75), HbTPS29 to HbTPS32 were predicted in the K04120 pathway (EC: 5.5.1.13), HbTPS33 to HbTPS41 were predicted in the K04121 pathway (EC: 4.2.3.19), HbTPS47 was predicted in the K17982 pathway (EC: 4.2.3.144), HbTPS18 and HbTPS19 were predicted in the K12742 pathway (EC: 4.2.3.27). The reaction map of the secondary metabolite pathways predicted by HbTPSs in the KEGG database was shown in Fig. 5. Prediction results information was shown in supplementary Table S1.

### Three-Dimensional Structure Prediction and Molecular Docking Analysis of HbTPSs

We selected one HbTPS per pathway and used AlphaFold2 to predict their three-dimensional structures based on their sequences. HbTPSs with different sequences could fold into different spatial structures and might have different functions. So 3D structure predictions and molecular docking analysis could reveal the functional sites of enzymes from the spatial structure. As could be seen from the quality assessment of the predicted results (supplementary file S4), the highest similarity of the seven protein sequences to their predicted template sequences reached more than 80%, and the lowest similarity was also 40%, and the average PLDDT was over 84, so the predicted structures were highly reliable and could be analyzed, and we selected the first-ranked result of each prediction for analysis.

The three-dimensional structures of the seven predicted HbTPSs were visualized in Fig. 6A, and all predicted HbTPSs had *N*-terminal helices. All HbTPSs contained  $\alpha\beta$  domain, while HbTPS30, HbTPS40, and HbTPS47 contained the  $\gamma$  domain in addition to the  $\alpha\beta$  domain. HbTPS30 was predicted to have the binding site DXDD of class II TPS in the  $\beta\gamma$  domain interface. HbTPS12, HbTPS17, HbTPS19, HbTPS40, HbTPS44, and HbTPS47 were predicted to have the binding



**Fig. 5** The reaction map of the seven secondary metabolite pathways predicted in the KEGG database for the HbTPSs

sites DDXXD and NSE/DTE of class I TPSs in the  $\alpha$  domain, and we also found the motif EA(X)W near the active site in the  $\alpha$  domain. The predicted three-dimensional structures of HbTPSs included  $\alpha\beta\gamma$  domains and  $\alpha\beta$  domains. HbTPS30, HbTPS40, and HbTPS47 contained  $\alpha\beta\gamma$  domains belonging to the TPS-c and TPS-e/f subfamilies clustered together. In contrast, HbTPS12, HbTPS17, HbTPS19, and HbTPS44 contained  $\alpha\beta$  domains belonging to TPS-a, TPS-b, and TPS-g subfamilies clustered together.

The predicted three-dimensional structures of HbTPSs were molecularly docked with the corresponding substrate using AutoDock, and the first-ranked docking result was selected for analysis. HbTPS12, HbTPS17, and HbTPS44 were predicted as sesquiterpene synthases in secondary metabolite pathway prediction, and their binding substrate was all FPP. HbTPS19 was predicted as isoprene synthase, and its binding substrate was DMAPP. HbTPS30 and HbTPS47 were predicted as diterpene synthases, and their binding substrates were GGPP. HbTPS40, which also was predicted as diterpene synthase, its binding substrate was ECP.

By observing the docking region of HbTPSs with the substrate, it can be seen (Fig. 6B) that the active sites of HbTPS12, HbTPS17, HbTPS19, HbTPS40, HbTPS44, and HbTPS47 were all in the  $\alpha$  domain, which was a hydrophobic pocket region formed by multiple  $\alpha$  helices. In the hydrophobic pocket region of HbTPS12, magnesium ion A formed coordination with N445, E453 (NSE/DTE), and FPP;

magnesium ions B and C formed coordination with D303 and D307 (DDXXD) and FPP; R456 and D458 formed hydrogen bonding force with FPP, which all indicated the interaction between conserved motifs and trinuclear magnesium ion cluster and substrate. This result was consistent with the catalytic mechanism of class I TPSs, and HbTPS17, HbTPS19, HbTPS40, HbTPS44, and HbTPS47 also showed similar docking as HbTPS12. It was worth noting that E393 in HbTPS17, E423 in HbTPS19, and E377 in HbTPS44 all generated hydrogen bonding forces with the diphosphate substrate, and these amino acid residues were all derived from motif EA(X)W, indicating that the motif EA(X)W was involved in the catalytic reaction. The active site of HbTPS30 was at the interface of the  $\beta\gamma$  domain, and its conserved motif was DXDD. Observing the binding map of HbTPS30 and its corresponding substrate GGPP, it can be seen that GGPP formed coordination with R343, R346, E479, and N431. R343, F335, and W339 had hydrogen bonding forces; R346 and E479 had hydrogen bonding forces; T427, F430, and N431 had hydrogen bonding forces; D385 and N431 had hydrogen bonding forces. This result was consistent with the catalytic mechanism of class II TPSs. Notably, the residues belonging to the conserved motif of DXDD in the docking results did not directly bind to the substrate GGPP.

### Analysis of Cis-Acting Elements of HbTPS Genes

Gene function analysis only showed the possible function and the participating metabolic path of the genes existing in the *H.brasiliensis*, but in fact, in a certain tissue, whether one gene can really play a role depended on its expression or not and its expression abundance, which could only be known by gene expression analysis. We conducted two aspects of gene expression analysis and its regulation, the first of which was that the upstream sequence of the gene could regulate the spatial and temporal expression of one gene. We analyzed 28 cis-acting elements involved in plant growth and development, stress responsiveness, and phytohormone responsiveness in HbTPS genes. The number of occurrences of cis-acting elements was shown in Fig. 7A. As shown in Fig. 7B, among the three major categories of cis-acting elements, the cis-acting elements related to phytohormone responsiveness had the highest percentage of occurrences overall. There were the least cis-acting elements in the members of TPS-e/f subfamily. As shown in Fig. 7C, among the cis-acting elements related to plant growth and development, AAGAA-motif and as-1 appeared more frequent; cis-acting elements related to stress responsiveness, ARE, STRE, and WUN-motif appeared more frequent; cis-acting elements related to phytohormone responsiveness, ABRE, GARE-motif, and ERE appeared more frequent. No cis-acting element was predicted in HbTPS7, HbTPS37, and HbTPS40 because the sequencing results of these data were poor.

### Differential Expression Analysis of HbTPSs

The other gene expression analysis was based on the transcriptome data. As shown in Fig. 8A, correlation analysis showed that the samples were well correlated,

**Fig. 6** **A** Three-dimensional structure of HbTPSs.  $\alpha$  domain was displayed in blue,  $\beta$  domain was displayed in green,  $\gamma$  domain was displayed in yellow, the *N*-terminal helix was displayed in magenta, the DDXXD conservative motif was shown in red, the NSE/DTE conservative motif was displayed in orange, the EA(X)W conservative motif was displayed with gray, and the DXDD conservative motif was shown with tint. **B** In the binding diagram of HbTPSs corresponding substrate and magnesium ion, the green ball was the magnesium ion, and the yellow dotted lines were hydrogen bond force (Color figure online)

and we had shown PCA figure of samples in supplemental file S4. By observing Fig. 8B, it could be seen that HbTPS16, HbTPS12, HbTPS13, and HbTPS11 were highly expressed in latex and clustered together. HbTPS4, HbTPS15, HbTPS5, and HbTPS24 were highly expressed in leaves and were clustered together. HbTPS3, HbTPS33, HbTPS22, HbTPS34, HbTPS17, HbTPS40 were also highly expressed in leaves and they were clustered together. In order to find as many differential genes as possible, we used  $\text{LogFC} = 2$ ,  $P\text{-value} < 0.05$  as the standard to make a volcano plot (Fig. 8C), it was found that HbTPS12, HbTPS13, HbTPS16, HbTPS11, and HbTPS6 were up-regulated in latex, except that the function of HbTPS16 was not annotated, the other four HbTPSs were (–)-germacrene D synthase in secondary metabolite pathways prediction. HbTPS3, HbTPS22, HbTPS17, HbTPS40, HbTPS34, HbTPS5, HbTPS20 were down-regulated in latex. Among them, HbTPS3 and HbTPS5 were annotated as (–)-germacrene D synthase, HbTPS17 was alpha-farnesene synthase, HbTPS20 was myrcene synthase, HbTPS22 was gamma-terpene synthase, and HbTPS34 and HbTPS40 were ent-kaurene synthases.

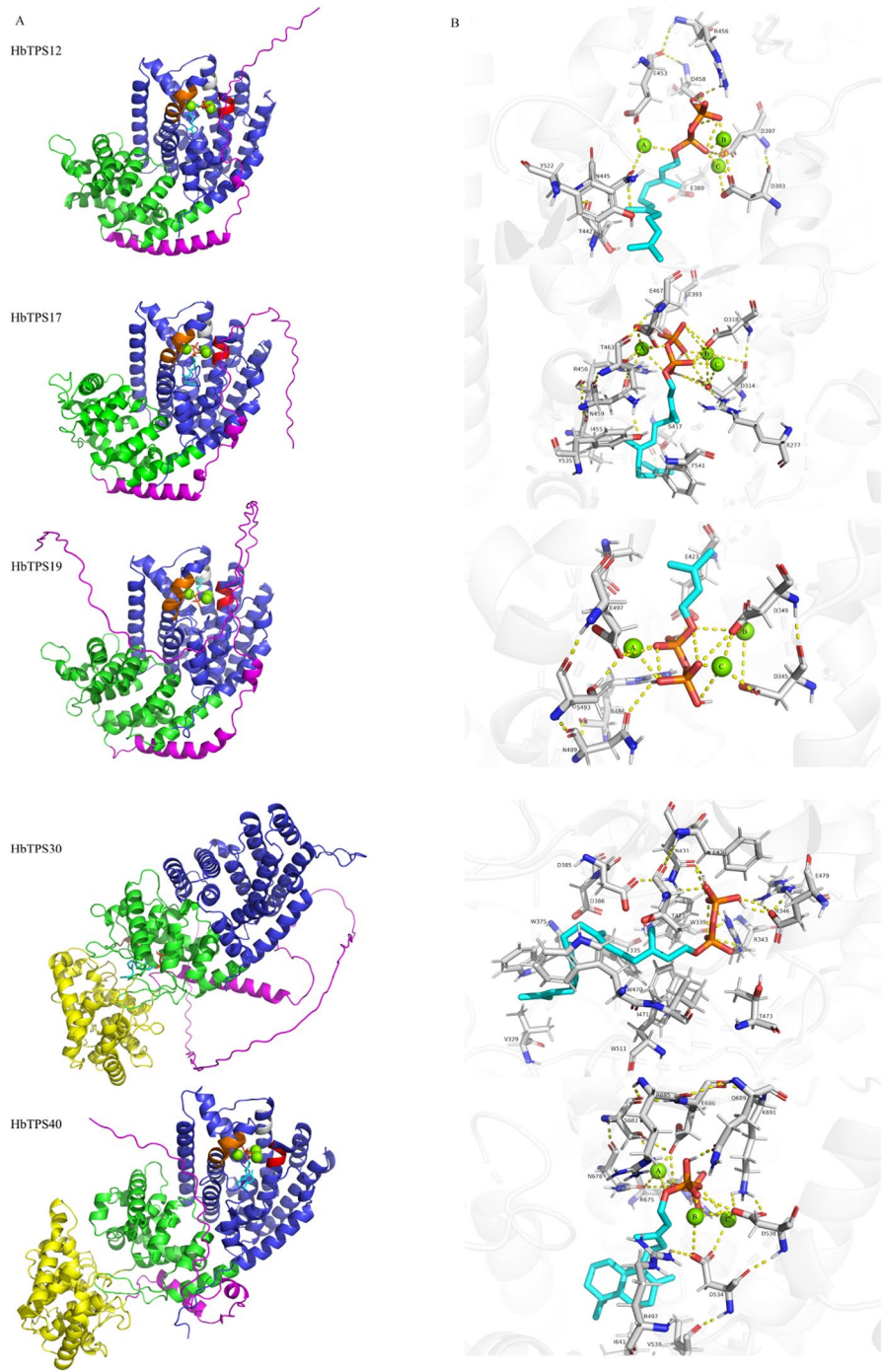
## Discussion

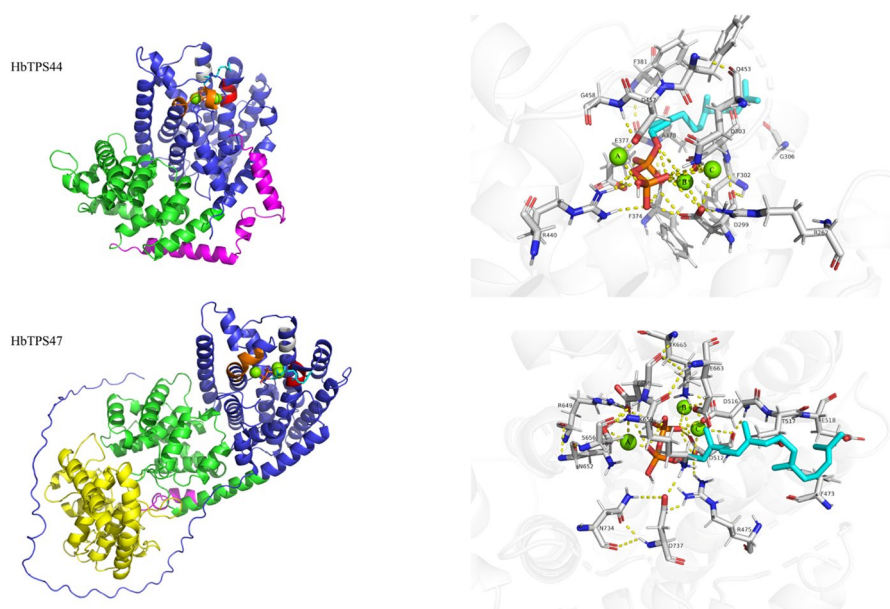
*H. brasiliensis* have significant economic value, which produces 90% of the world's natural rubber (Uthup et al. 2013). Besides this, there are other terpenoids such as cadinol, farnesol, squalene, etc. (Cornish 1993; Yamashita et al. 2016; Leme et al. 2020). TPSs can synthesize terpenoids such as monoterpenes, sesquiterpenes, diterpenes, and polyterpenes using terpene skeletons, which will affect the concentration of terpene skeleton monomers in the body, thus affecting the biosynthesis of natural rubber. Therefore, studying HbTPSs at the molecular level can help reveal the molecular mechanism of terpenoid synthesis in *H. brasiliensis*, laying the theoretical foundation for analyzing the regulatory mechanism of natural rubber synthesis in *H. brasiliensis*, and providing a theoretical basis for the molecular breeding and further development and utilization of *H. brasiliensis*.

In our study, 47 full-length HbTPSs were identified based on TPS's conserved *N*-terminal and *C*-terminal domains. The number of TPSs varied among plants (about 20–150), so the number of TPSs in *H. brasiliensis* was moderate compared to that of other plants (Chen et al. 2011).

In plants, TPS genes of TPS-c and TPS-e/f subfamilies contain 13–15 exons and 12–14 introns; TPS genes of TPS-a, TPS-b, and TPS-g subfamilies contain 6–8 exons and 5–7 introns (Bohlmann et al. 1998a; Trapp and Croteau 2001). In our study, there was a decrease in the number of exons and introns from HbTPS genes



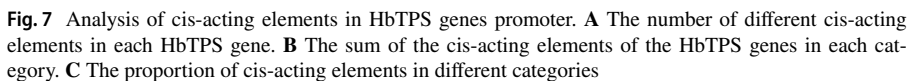


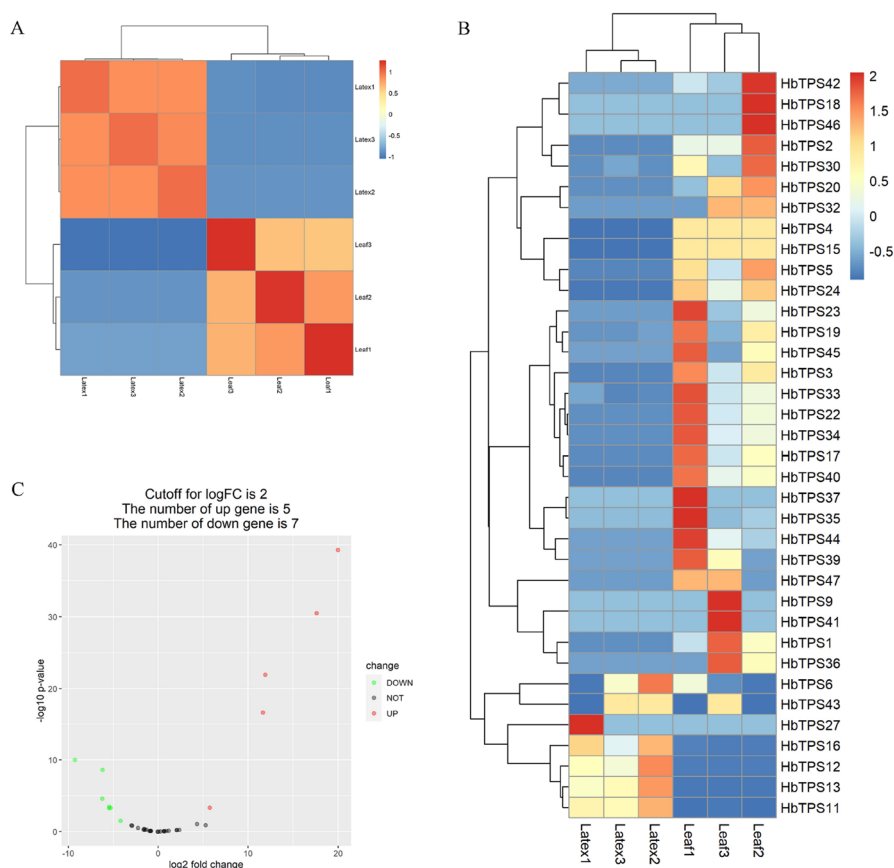


**Fig. 6** (continued)

of the subfamilies TPS-c and TPS-e/f to those of the subfamilies TPS-a, TPS-b, and TPS-g. This result was consistent with the idea that the exons and introns of TPS genes were gradually lost during evolution (Chen et al. 2021b). In our study, HbTPSs were divided into five subfamilies, TPS-a, TPS-b, TPS-c, TPS-e/f, and TPS-g. Most of the HbTPSs belonged to TPS-a and TPS-b subfamilies, which indicated that the HbTPSs of TPS-a and TPS-b subfamilies had been continuously expanded during evolution. TPSs of *H. brasiliensis* and *R. communis* were clustered together in each subfamily, which might be because they both belonged to the family Euphorbiaceae and were more closely related. In our study, HbTPS42–HbTPS46 belonged to TPS-g subfamily, which function was predicted as sesquiterpene synthase in secondary metabolite pathway prediction except for HbTPS42. This result differed from the previous studies of TPS-g subfamily genes encoding monoterpene synthases (Dudareva et al. 2003; Martin et al. 2010), and the exact functions of all these enzymes need to be further studied.

The results of functional annotation in InterPro and GO databases all indicated that the 47 HbTPSs mined were TPSs or had TPS functions, which indicated the reliability of the identification of HbTPSs. In the secondary metabolite pathway prediction results, HbTPS22 and HbTPS28 were gamma-terpinene synthases, and their binding substrate was GPP, which should be monoterpene synthases, but the exact pathway was not given in the annotation, so we could not determine whether they were indeed monoterpene synthases, which needs further experimental study. According to the relevant studies, three-dimensional structures of TPSs were essentially modular and could be composed of one, two, or three domains, so three-dimensional structures of TPSs mainly include  $\alpha$ ,  $\alpha\beta$ ,  $\beta\gamma$ , and  $\alpha\beta\gamma$ , while there are

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**Fig. 8** Latex3 represents the SRR15012153 sample, Latex2 represents the SRR15012154 sample, and Latex1 represents the SRR15012155 sample. Leaf3 represents the SRR15012156 sample, Leaf2 represents the SRR15012157 sample, and Leaf3 represents the SRR15012158 sample. **A** Sample correlation analysis. **B** Differential expression analysis of HbTPS genes. **C** Volcano map of HbTPSs differential expression genes

The  $\gamma$  domain of TPSs appeared to fall off during the evolution process (Köksal et al. 2011; Christianson 2017; Karunanithi and Zerbe 2019; Jia et al. 2022). In this study, HbTPS30, HbTPS40, and HbTPS47 belonged to TPS-c and TPS-e/f subfamilies that possessed  $\alpha\beta\gamma$  domains, while HbTPS12, HbTPS17, HbTPS19, and HbTPS44 belonged to TPS-a, TPS-b, and TPS-g subfamilies owned  $\alpha\beta$  domains, and it seemed that HbTPSs had lost their  $\gamma$  domains during the evolutionary process too. Notably, the HbTPS genes sequence analysis results also illustrated that during the evolutionary process, when TPSs showed monofunctionalization and loss of domains, it was also accompanied by the loss of introns and exons.

A previous paper suggested DDXXD and NSE/DTE were involved in divalent metal binding, and belonged to class I TPSs (Facchini and Chappell 1992; Christianson 2006; Zhou and Peters 2009). In our study, the active sites of HbTPS12,

HbTPS17, HbTPS19, HbTPS40, HbTPS44, and HbTPS47 were located in the  $\alpha$  domain. They had trinuclear magnesium ion cluster and conserved motifs DDXXD and NSE/DTE, which were the typical structures of class I TPSs. These HbTPSs might belong to class I TPSs. Notably, we also found conserved motif EA(X)W near the binding site of above HbTPSs, it might play an essential role in the catalytic reactions, and its specific function needs further research. Many studies had shown that DXDD had a role in initiating cyclization after protonation in class II TPSs (Kawaide et al. 2000; Hayashi et al. 2006; Kawaide 2006; Prisic et al. 2007). In our study, the active site of HbTPS30 was at the  $\beta\gamma$  domain interface, and it had DXDD conserved motif. It might belong to class II TPSs. No active sites and conserved motifs were found in the  $\gamma$  domain in all predicted three-dimensional structures of HbTPSs, which might be the reason for the loss of the  $\gamma$  domain during evolution.

Many studies have shown that TPS genes were involved in plant growth and development, responded to many biotic and abiotic stresses (McCormick et al. 2012; Tholl 2015; Huang et al. 2017; Campbell et al. 2018; Natalia et al. 2021), and played an essential role in stress resistance (Karunanithi et al. 2020; Yu et al. 2020; Zhou et al. 2020). The expression of TPS genes was regulated when plants were infested with pests or treated with phytohormones (Mariam et al. 2015; Li et al. 2018). In this study, a total of 873 cis-acting elements in 28 types related to plant growth and development, stress responsiveness, and phytohormone responsiveness were identified in the upstreams of HbTPS genes. These cis-acting elements might play essential roles in the growth and development, defense, and natural rubber production of *H.brasiliensis*. For example, AAGAA-motif and as-1 were necessary for growth and development. ARE, STRE, TC-rich repeats, and WUN-motif, which were related to stress. The abscisic acid response element (ABRE), MeJA response elements such as TGACG-motif, MYC, and CGTCA-motif, and ERE Ethylene Response Element were related to plant hormones. Among these cis-acting elements, the number of elements related to hormone response was the largest. It was speculated that the action of TPSs in plants was closely associated with the action of hormones, and this needs to be studied in depth by further experiments.

It has been shown that *H.brasiliensis* contained other terpenoids besides natural rubber (Cornish 1993; Tateyama et al. 1999; Phatthiya et al. 2007; Van Beilen and Poirier 2007; Yamashita et al. 2016; Leme et al. 2020). HbTPSs could directly catalyze the biosynthesis of these non-natural rubber terpenoids using IPP and its isomer DMAPP as substrates. The production of non-natural rubber terpenoids consumed IPPs, which probably reduced the yield of natural rubber (Chow et al. 2012; Li et al. 2015). It has been reported that the disruption of the squalene synthase gene in the isoprenoid branch pathway might significantly increase the substrate flow of natural rubber biosynthesis (Steinbüchel 2003; Men et al. 2018). In this study, HbTPS3, HbTPS22, HbTPS17, HbTPS40, HbTPS34, HbTPS5, HbTPS20 were down-regulated in expression in latex than that in leaves, which indicated that the catalytic products of these HbTPSs could be reduced, and the IPP flow to these non-natural rubber terpenoids probably decreased, which might lead to promote the production



of natural rubber. In the future, how these HbTPSs regulate the production of natural rubber in *H.brasiliensis* will be further studied by experimentation.

## Conclusion

In this study, we mined and screened genome-wide TPSs in *H.brasiliensis* and finally identified 47 full-length TPSs. Conserved motif analysis found that HbTPSs included RR(X)8 W, RLG, FRLLR, LY(X)AS, D(X)N(X)2Q, RDR, DDXXD, DXDD, RWD, EA(X)W, NSE/DTE 11 conserved motifs. Phylogenetic tree analysis divided HbTPSs into five subfamilies: TPS-a, TPS-b, TPS-c, TPS-e/f, and TPS-g. Functional annotation and secondary metabolite pathway prediction classified HbTPSs into sesquiterpene synthase, diterpene synthase, and isoprene synthase, which were grouped into seven pathways, K14173, K14175, K15803, K04120, K04121, K17982, and K12742. The three-dimensional structure prediction and molecular docking analysis of the seven HbTPSs of the seven pathways revealed the three-dimensional structure of HbTPSs and their binding interactions with the corresponding substrates and metal ions. Cis-acting element analysis of HbTPS genes identified 28 species of 873 cis-acting elements involved in plant growth and development, stress responsiveness, and phytohormone responsiveness. Differential expression analysis indicated that HbTPS genes impact natural rubber synthesis. The results of this study help to reveal the molecular mechanism of HbTPSs and can provide some new insights into the effective regulation of natural rubber biosynthesis.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s10528-022-10311-7>.

**Acknowledgements** Not applicable.

**Author Contributions** JHL and MHF contributed to the study's conception and design. JHL, DW, XL, WWH, and CX contributed to the analysis of the data. JHL and MHF wrote the paper. MHF, HMZ, and QM revised the paper. All authors read and approved the final version of the manuscript.

**Funding** This work was funded by Guangzhou Municipal Science and Technology Project (Grant No. 201903010025) and Science and Technology Planning Project of Guangdong Province (Grant No. 2016A010105020).

**Data Availability** The RNA-seq can download from NCBI SRA database (<http://www.ncbi.nlm.nih.gov/sra/>) under the accession number PRJNA742874. The genome sequence and annotation file can download from NCBI Genome database (<http://www.ncbi.nlm.nih.gov/genome/>) under the accession number ASM165405v1. Other datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

## Declarations

**Conflict of interest** The authors declared that they have no conflicts of interest.



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